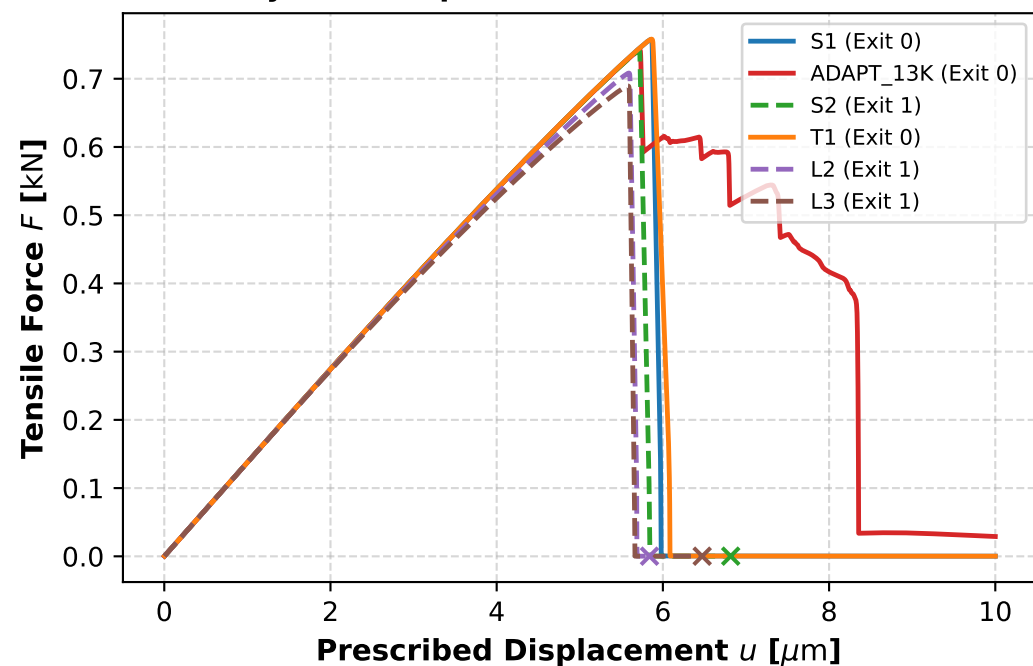
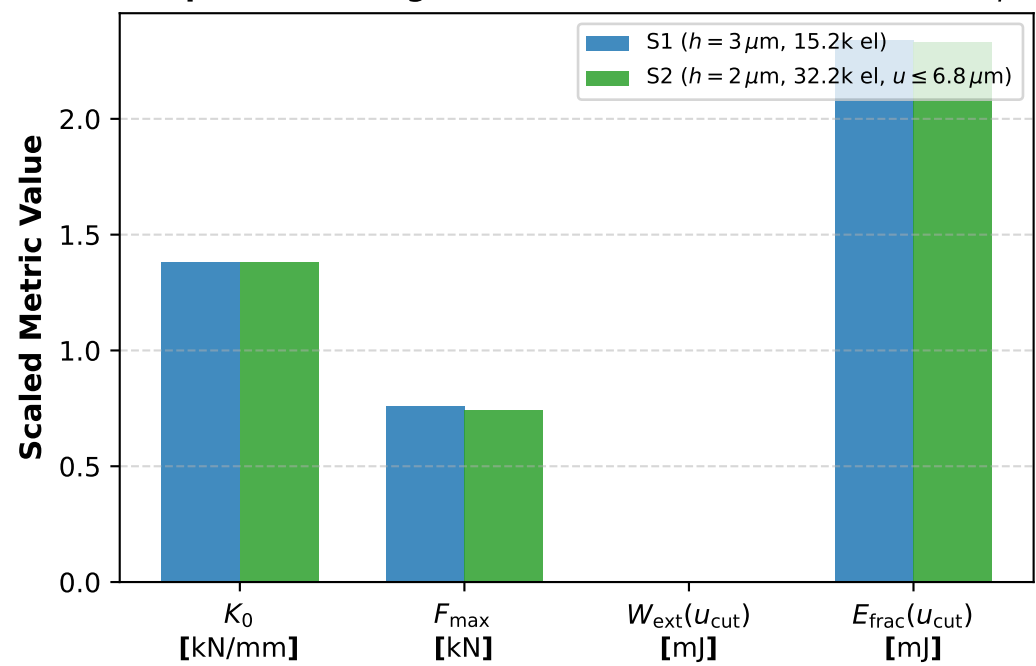
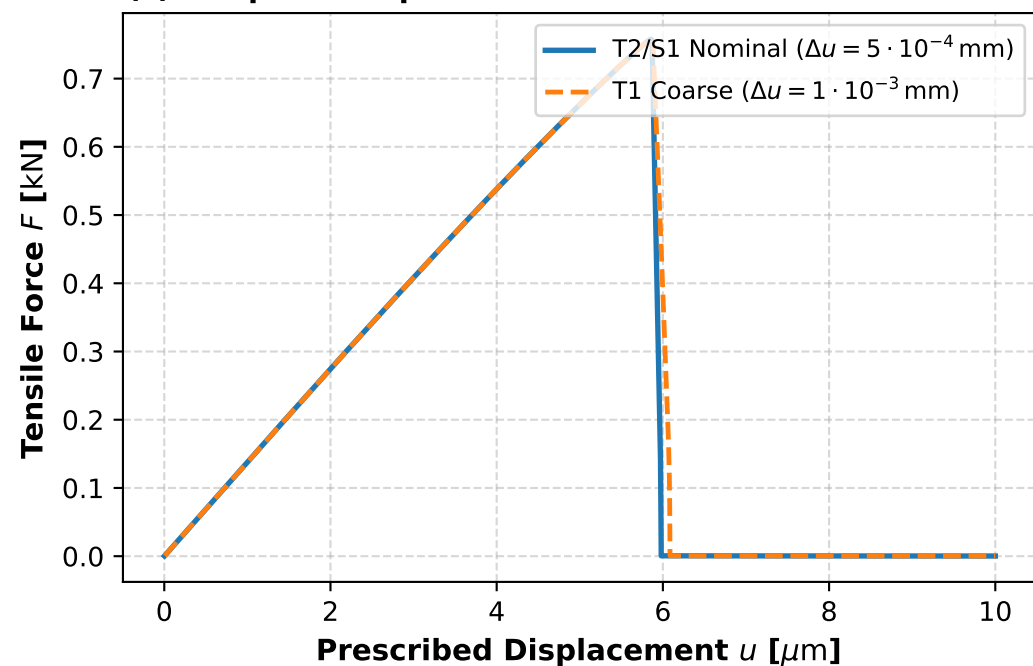
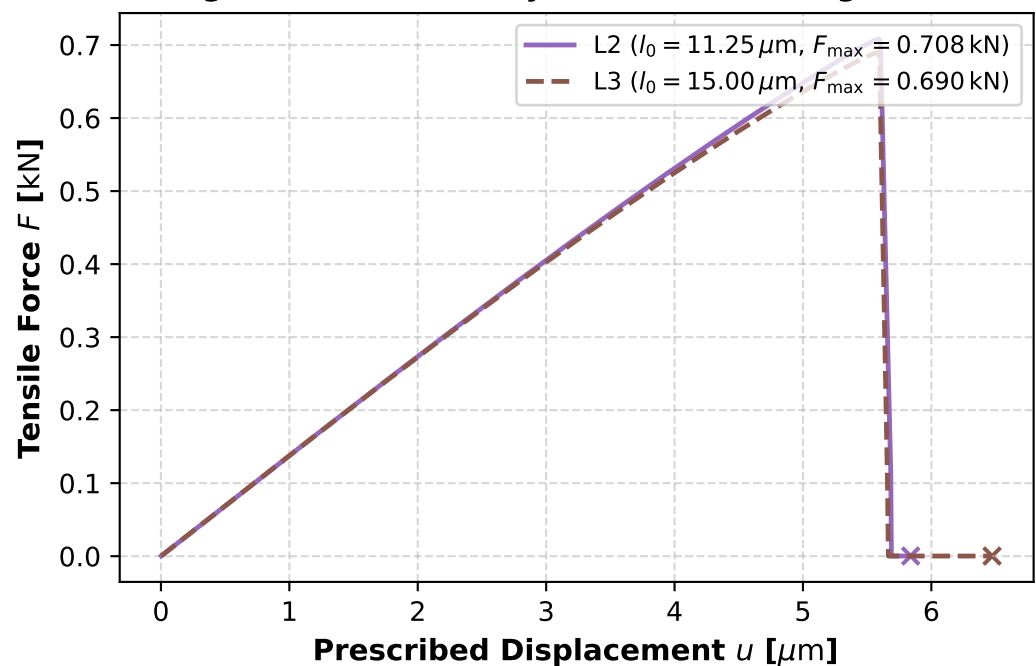
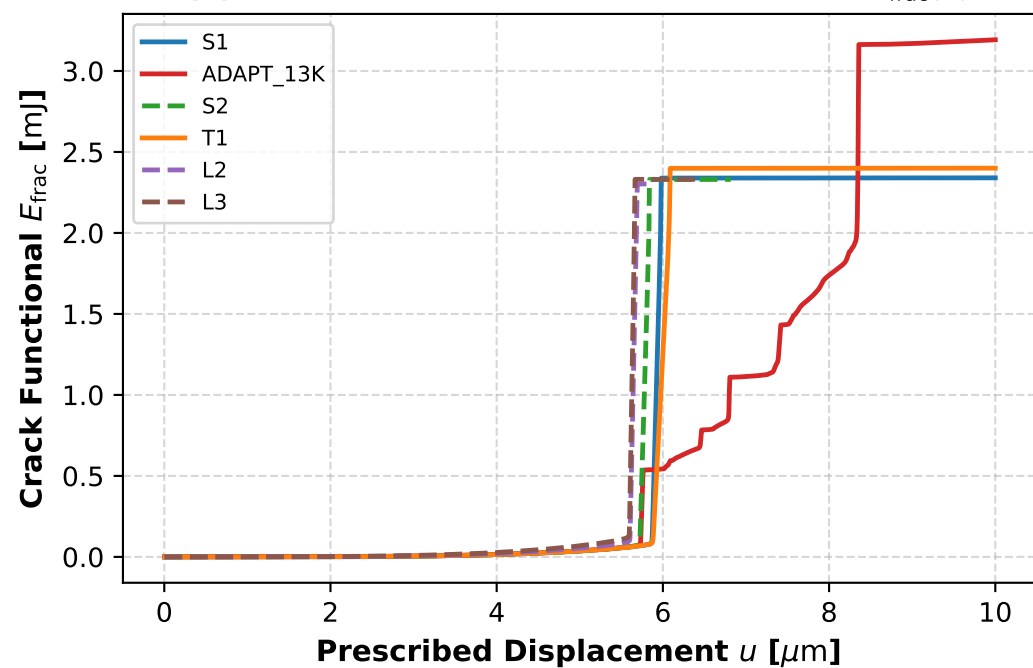


(a) Multi-Family Load-Displacement Curves (Truncations Marked ×)**(b) Spatial Convergence: S1 vs S2 at Matched $u = 6.82 \mu\text{m}$** **(c) Temporal Step Invariance: T1 Coarse vs T2 Nominal****(d) Length-Scale Sensitivity: Peak Load vs Regularization l_0** **(e) Phase-Field Crack Surface Functional $E_{\text{frac}}(u)$** **(f) Descriptive Energy Bookkeeping Diagnostic $\Delta_{\text{book}}(u)$** 